

AXLERO SOLUTIONS

ADVANCED DATA SCIENCE & MACHINE LEARNING ENGINEERING

12-WEEK PROJECT PROGRESS REPORT

Duration: 12 Weeks

Projects: 3 Advanced AI/ML Projects

Candidate: Sunil Kumar

PROJECT OVERVIEW

Project	Domain	Duration
Project 1 – OmniBrain	Agentic Multi-Modal RAG	Weeks 1–4
Project 2 – FedMed	Privacy-Preserving Federated Learning	Weeks 5–8
Project 3 – SwarmRL	Multi-Agent Deep Reinforcement Learning	Weeks 9–12

PROJECT 1 – OMNIBRAIN

Agentic Multi-Modal RAG Orchestrator

Duration: Week 1 – Week 4

Project Objective

Build an enterprise-grade Agentic Multi-Modal RAG platform capable of processing documents containing text, tables, images, charts, and structured data while producing grounded and cited responses through specialized AI agents.

Week 1 – Multi-Modal Document Processing

Activities Completed

- Project architecture setup
- FastAPI backend initialization
- React frontend initialization
- Document upload pipeline
- PDF processing pipeline
- Text extraction

- Image extraction
- Table extraction
- Document metadata generation
- Initial document chunking
- Qdrant vector database setup

Technologies

- Python
- FastAPI
- React
- Qdrant
- LangChain

Progress Outcome

A functional document ingestion pipeline capable of accepting enterprise documents and preparing text and visual content for downstream retrieval.

Week 2 – Agentic Architecture

Activities Completed

- LangGraph integration
- Supervisor Agent implementation
- Search Agent implementation
- Vision Agent foundation
- SQL Agent foundation
- Agent state management
- Agent routing logic
- Chat interface development
- Basic citation pipeline

Technologies

- LangGraph

- LangChain
- OpenAI / VLM
- React
- FastAPI

Progress Outcome

The multi-agent architecture was established, allowing user queries to be dynamically routed to appropriate specialized agents.

Week 3 – RAG and Self-Correction

Activities Completed

- Retrieval pipeline integration
- Semantic search
- Metadata filtering
- Context construction
- RAG response generation
- Citation generation
- Confidence scoring
- Self-correction workflow
- Irrelevant retrieval detection
- Query rewriting

Technologies

- Qdrant
- LangGraph
- LLM
- RAG Pipeline

Progress Outcome

The system became capable of generating grounded responses using retrieved context and performing retrieval refinement when initial results were insufficient.

Week 4 – Guardrails, Evaluation and UI Refinement

Activities Completed

- AI guardrails integration
- Prompt injection protection
- Hallucination monitoring
- Response validation
- Token usage tracking
- Latency monitoring
- Agent execution tracing
- Citation interface refinement
- Final dashboard improvements
- End-to-end testing

Technologies

- Langfuse
- NeMo Guardrails
- LangGraph
- React
- FastAPI

Final Project 1 Outcome

A complete Agentic Multi-Modal RAG prototype capable of:

- Multi-modal document understanding
- Agent-based reasoning
- Semantic retrieval
- Vision analysis
- Structured data querying
- Grounded responses
- Source citations
- AI safety and monitoring

PROJECT 2 – FEDMED

Cross-Silo Federated Learning Engine

Duration: Week 5 – Week 8

Project Objective

Develop a privacy-preserving federated learning platform allowing multiple hospitals to collaboratively train medical AI models without transferring raw patient data to a centralized server.

Week 5 – Medical AI Baseline

Activities Completed

- Federated learning architecture setup
- FastAPI backend setup
- Hospital node architecture
- Public MRI dataset preparation
- MRI preprocessing
- 3D U-Net implementation
- MONAI integration
- Centralized baseline training
- Initial segmentation evaluation

Technologies

- PyTorch
- MONAI
- FastAPI
- Medical Imaging Dataset

Progress Outcome

A centralized medical image segmentation baseline was established for comparison with the future federated model.

Week 6 – Federated Learning Infrastructure

Activities Completed

- Flower server setup
- Hospital Node A
- Hospital Node B
- Hospital Node C
- Dataset partitioning
- Federated training loop
- Model broadcasting
- Local model training
- FedAvg aggregation
- Client communication

Technologies

- Flower
- PyTorch
- FastAPI
- gRPC

Progress Outcome

Three independent hospital nodes were able to participate in collaborative model training without exchanging raw medical datasets.

Week 7 – Privacy and Secure Communication

Activities Completed

- TLS configuration
- Secure gRPC communication
- TenSEAL integration
- Homomorphic encryption
- Encrypted model updates

- Secure aggregation
- Differential privacy integration
- Privacy parameter management
- Node failure handling

Technologies

- TenSEAL
- gRPC
- TLS
- Differential Privacy
- Flower

Progress Outcome

The federated training pipeline was enhanced with privacy-preserving techniques, ensuring that model updates could be protected during distributed training.

Week 8 – Monitoring and Healthcare Dashboard

Activities Completed

- React dashboard
- Federated round monitoring
- Global accuracy visualization
- Training loss visualization
- Dice score visualization
- Hospital node status
- Encryption status
- Live metrics
- MRI segmentation visualization
- Final system testing

Technologies

- React

- TypeScript
- Recharts
- WebSockets
- PyTorch

Final Project 2 Outcome

A privacy-first federated healthcare AI platform capable of:

- Distributed medical model training
 - Local hospital training
 - Secure model aggregation
 - Encrypted model updates
 - Differential privacy
 - Real-time training monitoring
 - Medical segmentation visualization
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PROJECT 3 – SWARMRL

Multi-Agent Deep Reinforcement Learning Simulator

Duration: Week 9 – Week 12

Project Objective

Develop a multi-agent reinforcement learning simulator capable of training autonomous drone swarms to collaboratively explore disaster environments while minimizing collisions and maximizing coverage.

Week 9 – Simulation Environment

Activities Completed

- Project architecture setup
- PettingZoo/Gymnasium integration
- 3D environment creation
- Drone agent implementation
- Agent state management

- Continuous action space
- Observation space
- Environment boundaries
- Initial physics system

Technologies

- Python
- PettingZoo
- Gymnasium
- PyTorch

Progress Outcome

A functioning multi-agent environment was created with configurable autonomous drone agents and continuous movement.

Week 10 – Reward Engineering and MAPPO

Activities Completed

- Reward engine
- Exploration reward
- Collision penalty
- Boundary penalty
- Cooperation reward
- Coverage tracking
- Collision detection
- Ray RLlib integration
- MAPPO training configuration
- Actor network
- Centralized critic

Technologies

- Ray RLlib

- PyTorch
- MAPPO
- PettingZoo

Progress Outcome

The swarm became capable of learning cooperative behavior through multi-agent reinforcement learning.

Week 11 – Real-Time 3D Visualization

Activities Completed

- FastAPI simulation API
- WebSocket state streaming
- React dashboard
- Three.js integration
- React Three Fiber integration
- Real-time drone visualization
- Flight path visualization
- Sensor cone visualization
- Coverage visualization
- Dynamic obstacle support

Technologies

- React
- Three.js
- React Three Fiber
- FastAPI
- WebSockets

Progress Outcome

A real-time 3D visualization system was integrated, allowing users to observe swarm behavior and simulation state interactively.

Week 12 – Optimization, Analytics and Final Validation

Activities Completed

- Curriculum learning
- Wind simulation
- Training checkpointing
- Model evaluation
- Coverage analytics
- Collision analytics
- Reward analytics
- Agent performance analysis
- Performance optimization
- Final testing
- Documentation
- Deployment preparation

Technologies

- Ray RLlib
- PyTorch
- React
- Three.js
- Recharts
- Docker

Final Project 3 Outcome

A complete multi-agent reinforcement learning simulator capable of:

- Training autonomous drone swarms
- Cooperative exploration
- Collision avoidance
- Dynamic environment handling
- Curriculum learning

- Real-time 3D visualization
- Training analytics
- Model checkpointing
- Performance evaluation

OVERALL 12-WEEK PROGRESS

Project Weeks Major Achievement

OmniBrain 1–4 Agentic Multi-Modal RAG platform

FedMed 5–8 Privacy-preserving Federated Learning system

SwarmRL 9–12 Multi-Agent RL 3D swarm simulator

TECHNOLOGIES EXPLORED

AI / ML

- Python
- PyTorch
- TensorFlow concepts
- Deep Learning
- Reinforcement Learning
- Multi-Agent Reinforcement Learning

Agentic AI

- LangChain
- LangGraph
- RAG
- Vision-Language Models
- AI Guardrails

Privacy-Preserving ML

- Federated Learning

- Flower
- Homomorphic Encryption
- TenSEAL
- Differential Privacy

Autonomous Systems

- MAPPO
- Ray RLlib
- PettingZoo
- Gymnasium
- Multi-Agent Simulation

Full Stack

- FastAPI
- React
- TypeScript
- Tailwind CSS
- WebSockets

Visualization

- Three.js
- React Three Fiber
- Recharts

Infrastructure

- Docker
- Docker Compose
- Git
- CI/CD concepts

OVERALL LEARNING OUTCOMES

During the 12-week project period, the following capabilities were developed:

- Advanced AI system architecture
- Agentic AI development
- Multi-Modal RAG implementation
- Federated Learning
- Privacy-Preserving Machine Learning
- Deep Learning for medical imaging
- Multi-Agent Reinforcement Learning
- 3D simulation development
- Real-time system development
- API development
- Frontend engineering
- Model evaluation
- AI system monitoring
- Docker-based deployment
- Production-oriented software development

FINAL OUTCOME

The 12-week project progression resulted in three advanced AI engineering systems covering three distinct areas:

OmniBrain

Agentic AI + Multi-Modal RAG



FedMed

Privacy-Preserving Federated Learning



SwarmRL

Multi-Agent Deep Reinforcement Learning

Together, these projects demonstrate practical experience in **Agentic AI, Generative AI, Privacy-Preserving ML, Deep Learning, Distributed Systems, Reinforcement Learning, 3D Simulation, and Full-Stack AI Engineering.**

FINAL STATUS

Overall Project Status:

Completed / Final Validation

Projects Completed:

3 / 3

Documentation:

Completed

Testing:

Completed / Final Validation

Deployment:

Preparation / Demonstration Ready